The Universe as a Non-Equilibrium Information Processing System: Cosmological Dynamics from Information Geometry

Mariano Panzano Caballé

January 15, 2026

Abstract

This paper formalizes the emergence of cosmological dynamics from informational constraints. We solve the 120-order-of-magnitude discrepancy of the cosmological constant by identifying $\Lambda \approx R_U^{-2}$ as a geometric invariant of a constrained Fisher-Rao manifold.

7. The Emergence of Complexity

Life and consciousness are modeled as self-referential information structures that optimize the dissipation of informational curvature. The Fibonacci-constrained state space provides the necessary topological scaffolding for the emergence of stable biological coding (DNA), suggesting that life is a fundamental feature of the universe's geometric evolution.

Conclusion

The universe is a sovereign system moving from an Initial Coherent State toward a Master Resonance.